

Research Paper Summary Submission

Name of Student: Angat Nayanbhai Shah

Enrollment Number: 202203103510097

Class / Semester: 7th Semester - C

Subject: Explainable Artificial Intelligence (XAI)

Signature of Student: *Shah.A.N*

Navigating the Maze of Explainable AI: A Systematic Approach to Evaluating Methods and Metrics

Lukas Klein, Carsten Lüth, Udo Schlegel, Till Bungert, Mennatallah El-Assady
NeurIPS 2024, Datasets and Benchmarks Track

The field of Explainable AI (XAI) has witnessed explosive growth, yet its evaluation remains fragmented and unreliable. Existing studies often examine a narrow set of methods, ignore critical design parameters such as model architecture or modality and rely on a handful of metrics that introduce bias. This inconsistency has led to contradictory findings, particularly for widely used methods like GradCAM, leaving practitioners uncertain about which explanation tools to trust.

To address this gap, the authors introduce **LATEC**, a large-scale benchmark designed to provide systematic and transparent evaluation. LATEC encompasses **17 prominent XAI methods** assessed with **20 well-established metrics** across multiple architectures (CNNs and Transformers) and diverse modalities, including 2D images, 3D point clouds and volumetric medical data. In total, **7,560 unique experimental combinations** were conducted, making this the most comprehensive benchmark in XAI research to date. Importantly, the study also conducts *meta-evaluation* of metrics themselves, analyzing their ranking behaviors to expose inconsistencies and propose a more reliable evaluation scheme.

The findings reveal striking metric disagreements that significantly alter rankings depending on metric choice. For example, Expected Gradients consistently emerges as a top performer for both **faithfulness** and **robustness**, yet prior studies had largely ignored it, underscoring how selective evaluation obscures effective approaches. LATEC also releases an unprecedented dataset of **326,000 saliency maps** and **378,000 metric scores**, enabling reproducibility and serving as a foundation for future research.

Despite its scale, the benchmark is currently vision-centric, with future work needed in NLP and multimodal domains. Nonetheless, the methodological rigor and transparency of LATEC set a new standard for XAI evaluation. From a personal perspective, this paper is transformative: it forces the community to confront methodological weaknesses, shifts the focus from fragmented experiments to robust benchmarking and highlights overlooked methods that may define the next wave of trustworthy AI. By providing a systematic roadmap, LATEC elevates the reliability of XAI research and directly impacts its adoption in critical domains.

Processing, Priming, and Probing: Human Interventions for Explainability Alignment

Kenza Amara

CLR 2025 Workshop on Bidirectional Human-AI Alignment (BiAlign)

Explainable AI (XAI) aims to make complex machine learning systems transparent, but it faces a fundamental challenge: aligning explanations so they are simultaneously faithful to the model's reasoning and meaningful for human users. While human-centered XAI has made progress in accessibility, it often compromises factual correctness, producing explanations that appear convincing but do not reflect the underlying model. This misalignment erodes trust and undermines AI adoption in sensitive decision-making contexts.

To tackle this, the paper introduces the concept of **explainability alignment**, defined as the equilibrium between model-centric faithfulness and human-centric usability. It proposes the **Processing, Priming and Probing (PPP) framework**, which categorizes human interventions across different stages of the explanation pipeline. **Processing** involves post-hoc refinement of explanations, such as transforming saliency maps into sparse or verbalized narratives to reduce cognitive load. **Priming** occurs before explanation generation, embedding prior knowledge into the training process or explanation method through domain rules, physics-informed constraints or logical structures. **Probing** extends beyond predefined pipelines, engaging humans interactively to iteratively test, validate and refine explanations using counterfactual data or targeted user feedback.

The framework makes two critical contributions. First, it formalizes a structured taxonomy of interventions, equipping researchers with a vocabulary to design human-in-the-loop XAI. Second, it highlights the central trade-off: explanations that are “too plausible to be true” risk favoring human satisfaction over faithfulness, creating a false sense of trust. By framing interventions along a continuum, PPP shows that alignment requires conscious, multi-level strategies rather than ad hoc modifications.

As a limitation, the work is primarily conceptual, lacking large-scale empirical validation. Future research could test PPP interventions in real-world domains such as healthcare or finance to measure their impact on trust and decision quality. Personally, I view this framework as an essential step forward: it reframes explainability not as a binary compromise but as a dynamic negotiation where human interventions guide AI explanations toward both faithfulness and interpretability. Its structured approach could become foundational in designing transparent, accountable and trustworthy AI systems.